E-Waste Registration & Recycling Target Calculator

Mini-Task | Computer Engineering, Third Year

1. Objective

This program registers electronics companies under India's E-Waste (Management) Rules, 2022, calculates their annual EPR (Extended Producer Responsibility) recycling target, looks up the applicable registration fee, and reports each company's compliance status for a chosen financial year.

2. Approach

- Step 1 (Registration Form): Stored 5 sample companies (name, entity type, annual e-waste generated in MT) drawn from the reference list — Dixon Technologies, Kaynes Technology, Foxconn India, Tata Electronics, and Centum.
- Step 2 (EPR Target Calculator): Target (MT) = Annual E-Waste Generated x Target %, where Target % is looked up from the financial-year slab (60% for 2023-24/2024-25, 70% for 2025-26/2026-27, 80% for 2027-28 onward).
- Step 3 (Registration Fee Lookup): The computed EPR target (MT) is matched against the five fee slabs (Rs 2,500 up to Rs 15,00,000) to determine the registration fee.
- Step 4 (Output Report): For each company, the program prints Name, Type, Annual Generated, EPR Target, Registration Fee, and Compliance Status ("Compliant" if Actual Recycled >= Target, else "Non-Compliant"), using a second input (actual recycled MT).

3. Program (Python)

Full source below (ewaste_calculator.py). Run with: python3 ewaste_calculator.py

```
"""
E-Waste Registration & Recycling Target Calculator
Based on India's E-Waste (Management) Rules, 2022 (CPCB FAQ reference).

Steps implemented:
  1. Registration Form  - store company data (name, entity type, annual e-waste generated)
  2. EPR Target Calculator - target(MT) = annual e-waste generated * target % (per FY slab)
  3. Registration Fee Lookup - fee(Rs) based on the target (MT) slab
  4. Output Report - per-company summary with compliance status
"""

from dataclasses import dataclass


# ---------------------------------------------------------------------------
# Step 1: Registration Form (sample data for 5 companies from the reference list)
# ---------------------------------------------------------------------------

@dataclass
class Company:
    name: str
    entity_type: str          # Manufacturer / Producer / Recycler / Refurbisher
    annual_ewaste_mt: float    # Annual e-waste generated/handled, in metric tonnes
    actual_recycled_mt: float  # Actual e-waste recycled this FY, in metric tonnes


SAMPLE_COMPANIES = [
    Company("Dixon Technologies", "Manufacturer", annual_ewaste_mt=45,    actual_recycled_mt=30),
    Company("Kaynes Technology",  "Manufacturer", annual_ewaste_mt=80,    actual_recycled_mt=60),
    Company("Foxconn India",      "Producer",     annual_ewaste_mt=6500,  actual_recycled_mt=5000),
    Company("Tata Electronics",   "Manufacturer", annual_ewaste_mt=1200,  actual_recycled_mt=1000),
    Company("Centum",             "Refurbisher",  annual_ewaste_mt=300,   actual_recycled_mt=250),
]


# ---------------------------------------------------------------------------
# Step 2: EPR Target Calculator
# ---------------------------------------------------------------------------
```

```
def get_target_percent(financial_year: str) -> float:
    """Return the EPR recycling target % for a given financial year."""
    fy_slabs = {
        "2023-24": 0.60, "2024-25": 0.60,
        "2025-26": 0.70, "2026-27": 0.70,
    }
    if financial_year in fy_slabs:
        return fy_slabs[financial_year]

    # "2027-28 onward" -> 80%
    try:
        start_year = int(financial_year.split("-")[0])
    except (ValueError, IndexError):
        raise ValueError(f"Unrecognised financial year format: {financial_year}")

    if start_year >= 2027:
        return 0.80
    raise ValueError(f"No target % defined for financial year: {financial_year}")


def calculate_epr_target(annual_ewaste_mt: float, financial_year: str) -> float:
    """Target (MT) = Annual E-Waste Generated x Target %"""
    return annual_ewaste_mt * get_target_percent(financial_year)


# ---------------------------------------------------------------------------
# Step 3: Registration Fee Lookup
# ---------------------------------------------------------------------------

def get_registration_fee(target_mt: float) -> int:
    """Registration fee (Rs) based on the annual recycling target slab (in MT)."""
    if target_mt < 50:
        return 2_500
    elif target_mt < 100:
        return 7_500
    elif target_mt < 1000:
        return 1_50_000
    elif target_mt < 5000:
        return 10_00_000
    else:
        return 15_00_000


def format_inr(amount: int) -> str:
    """Format an integer amount using Indian digit grouping, e.g. 1500000 -> 15,00,000"""
    s = str(amount)
    if len(s) <= 3:
        return s
    last3 = s[-3:]
    rest = s[:-3]
    parts = []
    while len(rest) > 2:
        parts.insert(0, rest[-2:])
        rest = rest[:-2]
    if rest:
        parts.insert(0, rest)
    return ",".join(parts) + "," + last3


# ---------------------------------------------------------------------------
# Step 4: Output Report
# ---------------------------------------------------------------------------

def build_report(companies, financial_year: str):
    rows = []
    for c in companies:
        target = calculate_epr_target(c.annual_ewaste_mt, financial_year)
        fee = get_registration_fee(target)
        status = "Compliant" if c.actual_recycled_mt >= target else "Non-Compliant"
        rows.append({
            "name": c.name,
            "type": c.entity_type,
            "annual_ewaste_mt": c.annual_ewaste_mt,
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            "target_pct": get_target_percent(financial_year) * 100,
            "epr_target_mt": round(target, 2),
            "actual_recycled_mt": c.actual_recycled_mt,
            "fee_rs": fee,
            "status": status,
        })
    return rows


def print_report(rows, financial_year: str):
    print("=" * 100)
    print(f"E-WASTE EPR TARGET & FEE REPORT  |  Financial Year: {financial_year}")
    print("=" * 100)
    header = f"{'Company':<20}{'Type':<14}{'Generated(MT)':<15}{'Target%':<9}{'EPR Target(MT)':<17}
{'Recycled(MT)':<15}{'Fee (Rs)':<15}{'Status':<15}"
    print(header)
    print("-" * 100)
    for r in rows:
        print(
            f"{r['name']:<20}{r['type']:<14}{r['annual_ewaste_mt']:<15}"
            f"{r['target_pct']:<9.0f}{r['epr_target_mt']:<17}{r['actual_recycled_mt']:<15}"
            f"{format_inr(r['fee_rs']):<15}{r['status']:<15}"
        )
    print("=" * 100)


if __name__ == "__main__":
    FY = "2025-26"   # Change this to test other slabs, e.g. "2023-24", "2027-28"
    report_rows = build_report(SAMPLE_COMPANIES, FY)
    print_report(report_rows, FY)
```

4. Sample Output — Financial Year 2025-26 (Target = 70%)

Company	Type	Generated (MT)	Target %	EPR Target (MT)	Recycled (MT)	Fee (Rs)	Status
Dixon Technologies	Manufacturer	45	70%	31.5	30	2,500	Non-Compliant
Kaynes Technology	Manufacturer	80	70%	56.0	60	7,500	Compliant
Foxconn India	Producer	6500	70%	4550.0	5000	10,00,000	Compliant
Tata Electronics	Manufacturer	1200	70%	840.0	1000	1,50,000	Compliant
Centum	Refurbisher	300	70%	210.0	250	1,50,000	Compliant

At 70% (FY 2025-26), Dixon Technologies falls short of its 31.5 MT target (recycled only 30 MT) and is Non-Compliant; the other four companies meet or exceed their targets.

5. Rule / Clause Basis for Each Calculation

The EPR target percentages (60% for FY2023-24/24-25, 70% for FY2025-26/26-27, 80% from FY2027-28 onward) are drawn from the phased recycling/recovery obligation schedule under the E-Waste (Management) Rules, 2022, as clarified in the CPCB FAQ on the Rules. The registration fee slabs (Rs 2,500 to Rs 15,00,000, tiered by annual e-waste quantity handled in MT) correspond to the fee structure prescribed for Producer/Manufacturer/Refurbisher/Recycler registration under the same Rules and FAQ. The compliance check (actual recycled MT versus the computed EPR target MT) reflects each entity's obligation to meet its annual EPR target and file returns showing fulfillment, as required under the Rules' EPR compliance and reporting provisions. Entity classification (Manufacturer/Producer/Recycler/Refurbisher) follows the definitions in the Rules that determine which category of obligations (collection, recycling, or refurbishment) applies to a given company. All figures used are illustrative sample data for demonstration, not the companies' actual reported e-waste volumes.

6. Deliverables Checklist

- Working program (Python console app) — ewaste_calculator.py
- Sample output for the 5 companies — Section 4 above
- Explanation of rules used — Section 5 above